(equation 1.27)

$$\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m}\frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V \psi$$

$$\frac{\partial}{\partial t}|\psi|^2 = \psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t}$$

$$= \frac{i\hbar}{2m}\left[\psi^* \frac{\partial^2 \psi}{\partial x^2} - \psi \frac{\partial^2 \psi^*}{\partial x^2}\right]$$

$$- \frac{i}{\hbar} V \left[|\psi|^2 - |\psi|^2\right]$$

(equation 2.31)

$$1 = \int_0^a |A|^2 \sin^2(\beta x)\, dx = \frac{|A|^2}{2}\int_0^a \left[1 - \cos(2\beta x)\right] dx$$

$$= \frac{|A|^2}{2}\left[x - \frac{\sin(2\beta x)}{2\beta}\right]_0^a = \frac{|A|^2}{2}\left[a - \frac{\sin(2\beta a)}{2\beta}\right]$$

$$\beta_n = \frac{n\pi}{a} \Rightarrow \sin(2\beta a) = \sin(2\pi n) = 0$$

$$\Rightarrow 1 = \frac{|A|^2}{2} a \Rightarrow \boxed{|A| = \sqrt{\frac{2}{a}}}$$

(Proof 2.33)

$$\int \psi_m(x)\, \psi_n^*(x)\, dx = \delta_{mn}$$

proof:

$$\int \psi_m \psi_n^*\, dx = \frac{2}{a}\int \sin\left(\frac{n\pi}{a}x\right)\sin\left(\frac{m\pi}{a}x\right) dx$$

$$= \frac{1}{a}\int_0^a \left[\cos\frac{(n-m)\pi}{a}x - \cos\frac{(n+m)\pi}{a}x\right] dx$$

case $n = m$:

$$\int \psi_m \psi_n^*\, dx = \frac{1}{a}\left[a - a\frac{\sin\left(\frac{(n+m)\pi}{a}a\right)}{(n+m)\pi}\right] = 1$$

case $n \neq m$

$$\int \psi_m \psi_n^* \, dx = \frac{1}{a}\left[\frac{a \sin\frac{(n-m)\pi}{a}\, a}{(n-m)\pi} - \frac{a \sin\frac{(n+m)\pi}{a}\, a}{(n+m)\pi}\right] = 0$$

QED

Example 2.2

$$\psi(x,0) = \begin{cases} Ax(a-x) & 0 \le x \le a \\ 0 & \text{otherwise} \end{cases}$$

$$1 = \int_0^a A^2 x^2 (a^2 - 2ax + x^2)\, dx = A^2\left[a^2\frac{a^3}{3} - 2a\frac{a^4}{4} + \frac{a^5}{5}\right]$$

$$= A^2 a^5 \frac{10 - 15 + 6}{30} \Rightarrow \boxed{A = \sqrt{\frac{30}{a^5}}}$$

$$C_n = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{n\pi}{a}x\right)\psi(x,0)\, dx$$

$$= \frac{2\sqrt{15}}{a^2}\int_0^a x \sin\left(\frac{n\pi}{a}x\right)dx - \frac{2\sqrt{15}}{a^3}\int_0^a x^2 \sin\left(\frac{n\pi}{a}x\right)dx$$

(I) (II)

$$(I) = -\frac{a^2}{n\pi}(-1)^n + \frac{a}{n\pi}\int_0^a \cos\left(\frac{n\pi}{a}x\right)dx = -\frac{a^2}{n\pi}(-1)^n$$

$$(II) = -\frac{a^3}{n\pi}(-1)^n + \frac{2a}{n\pi}\int_0^a x\cos\left(\frac{n\pi}{a}x\right)dx$$

$$= -\frac{a^3}{n\pi}(-1)^n + \frac{2a}{n\pi}\left[-\frac{a}{n\pi}\int_0^a \sin\left(\frac{n\pi}{a}x\right)dx\right]$$

$$= -\frac{a^3}{n\pi}(-1)^n - 2\left(\frac{a}{n\pi}\right)^2\left(-\frac{a}{n\pi}\right)\cos\left(\frac{n\pi}{a}x\right)\Big|_0^a = -\frac{a^3}{n\pi}(-1)^n + 2\left(\frac{a}{n\pi}\right)^3\left[(-1)^n - 1\right]$$

$$\Rightarrow C_n = -\frac{2\sqrt{15}}{n\pi}(-1)^n + \frac{2\sqrt{15}}{n\pi}(-1)^n + \frac{4\sqrt{15}}{(n\pi)^3}\left[1 - (-1)^n\right]$$

$$\Rightarrow C_n = \begin{cases} \frac{8\sqrt{15}}{(n\pi)^3} & n = \text{odd} \\ 0 & n = \text{even} \end{cases}$$

$$\psi(x,t) = \sum_{n=1}^{\infty} C_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) e^{-i\left(n^2\pi^2\hbar/2ma^2\right)t}$$

$$= \sum_{n=1,3,5,\ldots} \frac{8\sqrt{30}}{a(n\pi)^3} \sin\left(\frac{n\pi}{a}x\right) e^{-i\left(n^2\pi^2\hbar/2ma^2\right)t}$$

(equation 2.48 and further)

$$\hat{a}_- \hat{a}_+ = \frac{1}{2\hbar m\omega}\left[+i\hat{p} + m\omega x\right]\left[-i\hat{p} + m\omega x\right]$$

$$= \frac{1}{2\hbar m\omega}\left[\hat{p}^2 + (m\omega x)^2 - i m\omega(x\hat{p} - \hat{p}x)\right]$$

$$= \frac{1}{2\hbar m\omega}\left[\hat{p}^2 + (m\omega x)^2\right] - \frac{i}{2\hbar}[x, \hat{p}]$$

$$= \frac{1}{\hbar\omega}\hat{H} + \frac{1}{2}$$

$$a_+ a_- = \frac{1}{2m\hbar\omega}\left[-i\hat{p} + m\omega x\right]\left[+i\hat{p} + m\omega x\right]$$

$$= \frac{1}{2m\hbar\omega}\left[\hat{p}^2 + (m\omega x)^2 - i m\omega(\hat{p}x - x\hat{p})\right]$$

$$= \frac{1}{\hbar\omega}\hat{H} - \frac{1}{2}$$

$$\Rightarrow [a_-, a_+] = \left(\frac{1}{\hbar\omega}\hat{H} + \frac{1}{2}\right) - \left(\frac{1}{\hbar\omega}H - \frac{1}{2}\right) = 1$$

(equation 2.60)

$$\psi_0(x) = A\, e^{-\frac{m\omega}{2\hbar}x^2}$$

$$1 = |A|^2 \int_{-\infty}^{+\infty} e^{-\frac{m\omega}{\hbar}x^2} = |A|^2 \sqrt{\frac{\pi\hbar}{m\omega}}$$

$$\Rightarrow A = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}$$

Plugging ψ_0 into eq 2.58 gives

$$\hbar\omega\left(\hat{a}_+ \hat{a}_- + \frac{1}{2}\right)\psi_0 = \frac{\hbar\omega}{2}\psi_0 = E_0\psi_0 \qquad \text{(because } \hat{a}_-\psi_0 = 0\text{)}$$

$$\Rightarrow E_0 = \frac{\hbar\omega}{2}$$

Example 2.4

$$\psi_1(x) = A_1 \hat{a}_+ (\psi_0) = \frac{A_1}{\sqrt{2m\hbar\omega}} \left[-\hbar \frac{d}{dx} + m\omega x \right] \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$$

$$= \frac{A_1}{\sqrt{2\hbar m\omega}} \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} 2m\omega x \, e^{-\frac{m\omega}{2\hbar}x^2}$$

$$= A_1 \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \sqrt{\frac{2m\omega}{\hbar}} \, x \, e^{-\frac{m\omega}{2\hbar}x^2}$$

$$1 = |A_1|^2 \sqrt{\frac{m\omega}{\pi\hbar}} \frac{2m\omega}{\hbar} \int_{-\infty}^{+\infty} x^2 e^{-\frac{m\omega}{\hbar}x^2} dx$$

The back of the book tells us that $\int_0^{+\infty} x^2 e^{-\frac{x^2}{a^2}} = \sqrt{\pi}\, 2 \left(\frac{a}{2}\right)^3$

$$\Rightarrow 1 = |A_1|^2 \sqrt{\frac{m\omega}{\pi\hbar}} \frac{2m\omega}{\hbar} \cdot \frac{\sqrt{\pi}}{2} \left(\frac{\hbar}{m\omega} \right)^{3/2} = |A_1|^2$$

$$\Rightarrow A_1 = 1$$

(Equation 2.65 and further)

$$\int_{-\infty}^{+\infty} f^* (\hat{a}_\pm g) dx = \frac{1}{\sqrt{2m\hbar\omega}} \int_{-\infty}^{+\infty} f^* \left(\mp \hbar \frac{d}{dx} + m\omega x \right) g \, dx$$

$$\pm\hbar \int_{-\infty}^{+\infty} f^* \frac{dg}{dx} dx = \pm\hbar \left[f^* g \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} \frac{df^*}{dx} g \, dx \right]$$

$$\Rightarrow \int_{-\infty}^{+\infty} f^* (\hat{a}_\pm g) dx = \frac{1}{\sqrt{2m\hbar\omega}} \int_{-\infty}^{+\infty} \left[\mp\hbar \frac{df^*}{dx} g + m\omega x f^* g \right] dx$$

$$= \frac{1}{\sqrt{2m\hbar\omega}} \int_{-\infty}^{+\infty} g \left[\left(\mp\hbar \frac{\partial}{\partial x} + m\omega x \right) f \right]^* dx$$

$$= \int_{-\infty}^{+\infty} g \left[\hat{a}_\mp f \right]^* dx$$

Applying this to $\int_{-\infty}^{+\infty} (\hat{a}_{\pm}\psi_n)^* (\hat{a}_{\pm}\psi_n) dx$ gives

$$\int_{-\infty}^{+\infty} (\hat{a}_{\pm}\psi_n)^* (\hat{a}_{\pm}\psi_n) dx = \int_{-\infty}^{+\infty} (\hat{a}_{\mp}\hat{a}_{\pm}\psi_n)^* \psi_n dx$$

Equation (2.58) and (2.62) together give us

$$\hbar\omega\left(\hat{a}_+\hat{a}_- + \frac{1}{2}\right)\psi_n = \left(n+\frac{1}{2}\right)\hbar\omega\psi_n$$

$$\Rightarrow \hbar\omega\,\hat{a}_+\hat{a}_-\psi_n = n\hbar\omega\psi_n$$

$$\Rightarrow \hat{a}_+\hat{a}_-\psi_n = n\psi_n$$

Similarly:

$$\hbar\omega\left(\hat{a}_-\hat{a}_+ - \frac{1}{2}\right)\psi_n = \left(n+\frac{1}{2}\right)\hbar\omega\psi_n$$

$$\Rightarrow \hat{a}_-\hat{a}_+\psi_n = (n+1)\psi_n$$

And therefore

$$\int_{-\infty}^{+\infty} (\hat{a}_+\psi_n)^* (\hat{a}_+\psi_n) dx = \int_{-\infty}^{+\infty} (\hat{a}_-\hat{a}_+\psi_n)^* \psi_n dx$$

$$= (n+1)\int_{-\infty}^{+\infty} |\psi_n|^2 dx$$

But using 2.64, we also find

$$\int_{-\infty}^{+\infty} (\hat{a}_+\psi_n)^* (\hat{a}_+\psi_n) dx = |C_n|^2 \int_{-\infty}^{+\infty} |\psi_{n+1}|^2 dx$$

Because $\int_{-\infty}^{+\infty} |\psi_n|^2 dx = \int_{-\infty}^{+\infty} |\psi_{n+1}|^2 dx = 1$, we find

$$|C_n|^2 = n+1 \Rightarrow C_n = \sqrt{n+1}$$

Similarly,

$$\int_{-\infty}^{+\infty} (\hat{a}_- \psi_n)^* (\hat{a}_- \psi_n)\, dx = \int_{-\infty}^{+\infty} (\hat{a}_+ \hat{a}_- \psi_n)^* \psi_n\, dx$$

$$= n \int_{-\infty}^{+\infty} |\psi_n|^2\, dx = n$$

And also

$$\int_{-\infty}^{+\infty} (\hat{a}_- \psi_n)^* (\hat{a}_- \psi_n)\, dx = |d_n|^2 \int_{-\infty}^{+\infty} |\psi_{n-1}|^2\, dx = |d_n|^2$$

$$\Rightarrow |d_n|^2 = n \quad \Rightarrow d_n = \sqrt{n}$$

And finally

$$\hat{a}_+ \psi_n = \sqrt{n+1}\, \psi_{n+1} \quad \text{and} \quad \hat{a}_- \psi_n = \sqrt{n}\, \psi_{n-1}$$

From the left on we find: $\psi_{n+1} = \dfrac{\hat{a}_+ \psi_n}{\sqrt{n+1}}$

So that

$$\psi_1 = \hat{a}_+ \psi_0$$

$$\psi_2 = \frac{\hat{a}_+ \psi_1}{\sqrt{2}} = \frac{(\hat{a}_+)^2 \psi_0}{\sqrt{2}}$$

$$\psi_3 = \frac{1}{\sqrt{3}} \hat{a}_+ \psi_2 = \frac{1}{\sqrt{3!}} (\hat{a}_+)^3 \psi_0 \quad \text{and so on.}$$

$$\Rightarrow \psi_n = \frac{1}{\sqrt{n!}} (\hat{a}_+)^n \psi_0$$

Claim: $\int_{-\infty}^{+\infty} \psi_m^* \psi_n \, dx = \delta_{mn}$

Proof:

$$\int_{-\infty}^{+\infty} \psi_m^* \psi_n \, dx \overset{(2.66)}{=} \frac{1}{n} \int_{-\infty}^{\infty} \psi_m^* \left(\hat{a}_+ \hat{a}_- \right) \psi_n \, dx$$

$$\overset{(2.65)}{=} \frac{1}{n} \int_{-\infty}^{+\infty} \left(\hat{a}_- \psi_m \right)^* \left(\hat{a}_- \psi_n \right) dx \overset{(2.65)}{=} \frac{1}{n} \int_{-\infty}^{+\infty} \left(\hat{a}_+ \hat{a}_- \psi_m \right)^* \psi_n \, dx$$

$$= \frac{m}{n} \int_{-\infty}^{+\infty} \psi_m^* \psi_n \, dx$$

If $m \neq n$, this equality can only hold if

$$\int_{-\infty}^{+\infty} \psi_m^* \psi_n \, dx = 0$$

We know all the ψ_k are normalised.

Q.E.D

Example 2.5

$$\langle V \rangle_n = \left\langle \frac{1}{2} m\omega^2 x^2 \right\rangle_n = \frac{1}{2} m\omega^2 \int_{-\infty}^{+\infty} \psi_n^* x^2 \psi_n \, dx$$

$$\hat{a}_\pm = \frac{1}{\sqrt{2m\hbar\omega}} \left(\mp i\hat{p} + m\omega x \right)$$

$$\Rightarrow \hat{a}_+ + \hat{a}_- = \frac{1}{\sqrt{2m\hbar\omega}} \left[-i\hat{p} + m\omega x + i\hat{p} + m\omega x \right]$$

$$\Leftrightarrow \hat{a}_+ + \hat{a}_- = \frac{1}{\sqrt{2m\hbar\omega}} 2m\omega x = \sqrt{\frac{2m\omega}{\hbar}} x$$

$$\Rightarrow \boxed{x = \sqrt{\frac{\hbar}{2m\omega}} \left(\hat{a}_+ + \hat{a}_- \right)}$$

Analogously,

$$\hat{a}_+ - \hat{a}_- = \frac{1}{\sqrt{2m\hbar\omega}} (-2i) \hat{p}$$

$$\Rightarrow \boxed{\hat{p} = i\sqrt{\frac{m\hbar\omega}{2}} \left(\hat{a}_+ - \hat{a}_- \right)}$$

Therefore,

$$x^2 = \frac{\hbar}{2m\omega} \left[(\hat{a}_+)^2 + (\hat{a}_+ \hat{a}_-) + (\hat{a}_- \hat{a}_+) + (\hat{a}_-)^2 \right]$$

$$\Rightarrow \langle V \rangle_n = \frac{\hbar\omega}{4} \int_{-\infty}^{+\infty} \psi_n^* \left[\underbrace{(\hat{a}_+)^2}_{\text{(I)}} + \underbrace{(\hat{a}_+ \hat{a}_-)}_{\text{(II)}} + \underbrace{(\hat{a}_- \hat{a}_+)}_{\text{(III)}} + \underbrace{(\hat{a}_-)^2}_{\text{IV}} \right] \psi_n \, dx$$

(I) : $(\hat{a}_+)^2 \psi_n \sim \psi_{n+2}$, and the integral $\int_{-\infty}^{+\infty} \psi_n^* \psi_{n+2} \, dx = 0$

So this term vanishes.

IV: The exact same argument shows this also vanishes.

$$\Rightarrow \langle V \rangle_n = \frac{\hbar\omega}{4} \int_{-\infty}^{+\infty} \psi_n^* \left[\sqrt{n}\cdot\sqrt{n}\,\psi_n + \sqrt{n+1}\sqrt{n+1}\,\psi_n \right] dx$$

$$= \frac{\hbar\omega}{4}\left(n + n+1\right)$$

$$= \frac{1}{2}\hbar\omega\left(n+\frac{1}{2}\right)$$

2.3.2

$$\xi = \sqrt{\frac{m\omega}{\hbar}}\, x \quad \Rightarrow \frac{d}{dx} = \frac{d\xi}{dx}\frac{d}{d\xi} = \sqrt{\frac{m\omega}{\hbar}}\frac{d}{d\xi}$$

Therefore

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi = E\psi$$

$$\Leftrightarrow \frac{m\omega}{\hbar}\frac{d^2\psi}{d\xi^2} = \frac{m\omega}{\hbar}\xi^2\psi - \frac{2mE}{\hbar^2}\psi$$

$$\Leftrightarrow \frac{d^2\psi}{d\xi^2} = \left[\xi^2 - K\right]\psi \quad (*)$$

For $\xi >> K$: $\frac{d^2\psi}{d\xi^2} \approx \xi^2\psi \quad (**)$

Guessing an exponential form : $\psi(\xi) \approx e^{g(\xi)}$ with some function g.

$$\Rightarrow \psi' \approx g' e^g$$

$$\Rightarrow \psi'' \approx \left[g'' + (g')^2\right]\psi$$

To obey $(**)$, we need $g' \sim \pm\xi$, and so $g'' \sim \pm 1$

so that for large ξ, we have $g'' + (g')^2 \approx (g')^2$

We find $g' \sim \pm\xi \Rightarrow g \sim \pm\frac{\xi^2}{2}$

That is why it makes sense to suggest

$$\psi(\xi) \approx A e^{-\frac{\xi^2}{2}} + B e^{\frac{\xi^2}{2}}$$

The B term is not normalisable $\Rightarrow B = 0$

$$\Rightarrow \psi(\xi) \sim A e^{-\frac{\xi^2}{2}}$$

We estimate that for large ξ, $\psi(\xi)$ is dominated by $e^{-\frac{\xi^2}{2}}$

which is why we suggest for (*) a solution of the form

$$\psi(\xi) = h(\xi)\, e^{-\frac{\xi^2}{2}} \qquad (***)$$

$$\Rightarrow \psi' = \left[h' - h\xi\right] e^{-\frac{\xi^2}{2}}$$

$$\Rightarrow \psi'' = \left[h'' - 2h'\xi - h + h\xi^2\right] e^{-\frac{\xi^2}{2}}$$

Plugging this into (*):

$$h'' - 2\xi h' + (\xi^2 - 1)h = \left[\xi^2 - K\right] h$$

$$\Leftrightarrow h'' - 2\xi h' + [K-1]\, h = 0$$

Trying $h = \sum_{j=0}^{\infty} a_j \xi^j$

$$\Rightarrow h' = \sum_{j=0}^{\infty} j \cdot a_j \xi^{j-1} = \sum_{j=0}^{\infty} (j+1)\, a_{j+1} \xi^j$$

$$\Rightarrow h'' = \sum_{j=0}^{\infty} j(j+1)\, a_{j+1} \xi^{j-1} = \sum_{j=0}^{\infty} (j+1)(j+2)\, a_{j+2} \xi^j$$

$$\Rightarrow \sum_{j=0}^{\infty} (j+1)(j+2)\, a_{j+2} \xi^j - \sum_{j=0}^{\infty} 2(j+1)\, a_{j+1} \xi^{j+1} + \sum_{j=0}^{\infty} [K-1]\, a_j \xi^j = 0$$

This can only be 0 for any ξ if each of the terms adds up to 0:

$$(j+1)(j+2)\, a_{j+2} - 2j a_j + (K-1) a_j = 0$$

$$\Rightarrow a_{j+2} = \frac{2j + 1 - K}{(j+1)(j+2)}\, a_j \qquad (\text{for } j = 0, 1, 2, \ldots)$$

For large j: $a_{j+2} \approx \frac{2}{j} a_j$

Case $j = 2m$: $a_{2m+2} \approx \frac{2}{2m} a_{2m} \approx \frac{1}{m} a_{2m} \approx \frac{1}{m} \frac{1}{(m-1)} a_{2m-2} \cdots$

$\Rightarrow a_{j+2} = a_{2m+2} \sim \frac{c}{m!} = \frac{c}{(j/2)!}$

$$\Rightarrow h_{even}(\xi) = \sum_{\substack{j=0 \\ j \text{ even}}}^{\infty} a_j \xi^j \sim c \sum_{\substack{j=0 \\ j \text{ even}}}^{\infty} \frac{1}{(j/2)!} \xi^j = \sum_{k=0}^{\infty} \frac{1}{k!} \xi^{2k}$$

$$= e^{\xi^2}$$

Plugging this into (***), we find

$$\psi(\xi) \sim e^{\xi^2} \cdot e^{-\frac{\xi^2}{2}}$$

which is not normalisable. We find that $h_{even}(\xi)$ must terminate at some a_i. There must therefore be some N such that

$$a_{N+2} = \frac{2N+1-K}{(N+1)(N+2)} a_N = 0$$

$$\Rightarrow N = \frac{K}{2} - \frac{1}{2} = \frac{E}{\hbar\omega} - \frac{1}{2} \quad \Leftrightarrow E = \hbar\omega\left(N + \frac{1}{2}\right)$$

This shows that for every n even, there is a solution for (*) with energy $E_n = \hbar\omega(n + \frac{1}{2})$ (with $n = 0, 2, 4, \ldots$)

Case $j = 2m+1$:

$$a_{j+2} = \frac{2j+1-K}{(j+1)(j+2)} a_j$$

and it is still true for large j that $a_{j+2} \approx \frac{2}{j} a_j$

$$a_{2m+3} \approx \frac{2}{2m+1} a_{2m+1} \approx \frac{1}{m} a_{2m+1}$$

$$\Rightarrow \quad a_{2m+3} \approx \frac{1}{m} a_{2m+1} \approx \frac{1}{m}\frac{1}{m-1} a_{2m-1} \approx \cdots$$

$$\Rightarrow \quad a_{j+2} = a_{2m+3} \sim \frac{C'}{m!} = \frac{C'}{\left(\frac{j}{2} - \frac{1}{2}\right)!}$$

We find

$$h_{odd}(\xi) = \sum_{\substack{j=1 \\ j \text{ odd}}}^{\infty} a_j \xi^j \approx C' \sum_{\substack{j=1 \\ j \text{ odd}}}^{\infty} \frac{1}{\left(\frac{j}{2} - \frac{1}{2}\right)!} \xi^j$$

$$= C' \sum_{k=0}^{\infty} \frac{1}{k!} \xi^{2k+1} = C' \xi e^{\xi^2}$$

Plugging this into (***) gives

$$\psi(\xi) \sim \xi e^{\xi^2} e^{-\frac{\xi^2}{2}}$$

which is again not normalisable. Therefore h_{odd} must also terminate, and the same conclusion is reached as for h_{even}.

Taking both cases together tells us that has to either obey $K = 2N+1$ with N even, with h_{odd} being 0 from the start. Or $K = 2N+1$ with N odd, with h_{even} equal to 0 from the start.

The conclusion is that the allowed values for the energy are

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right) \qquad (n = 0, 1, 2, 3, \ldots)$$

For the allowed values of $K = 2n+1$:

$$a_{j+2} = \frac{2j+1-K}{(j+1)(j+2)} a_j = \frac{2(j-n)}{(j+1)(j+2)} a_j$$

$n=0$

$h_{odd}(\xi)=0$ and $h_{even}(\xi)=a_0$

$n=1$

$h_{even}(\xi)=0$ and $h_{odd}=a_1\xi$ $\left(\text{so that } \psi(\xi)=a_1\xi e^{-\frac{\xi^2}{2}}\right)$

$n=2$

$h_{odd}(\xi)=0$ and $h_{even}(\xi)=a_0+a_2\xi^2$ with $a_2=-2a_0$

$$\Rightarrow h(\xi)=a_0(1-2\xi^2)$$

Example 2.6

$$\psi(x,0)=\begin{cases}A & -a<x<a\\ 0 & \text{otherwise}\end{cases}$$

$$1=|A|^2\,2a \Rightarrow A=\frac{1}{\sqrt{2a}}$$

$$\Phi(k)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}\psi(x,0)\,e^{-ikx}\,dx$$

$$=\frac{1}{2\sqrt{\pi a}}\int_{-a}^{a}e^{-ikx}\,dx \qquad -2i\sin ka$$

$$=\frac{1}{2\sqrt{\pi a}}\,\frac{e^{-ikx}}{-ik}\Big|_{-a}^{a}$$

$$=\frac{i}{2k\sqrt{\pi a}}\left(e^{-ika}-e^{ika}\right)$$

$$=\frac{\sin(ka)}{k\sqrt{\pi a}}$$

We find

$$\psi(x,t)=\frac{1}{\pi\sqrt{2a}}\int_{-\infty}^{+\infty}\frac{\sin(ka)}{k}\,e^{i\left(kx-\frac{\hbar k^2}{2m}t\right)}\,dk$$

Derivation of 2.140 to 2.142

$$\begin{cases} F+G = A+B & (2.136) \\ F-G = A(1+2i\beta)-B(1-2i\beta) & (2.138) \end{cases} \quad \text{and} \quad G=0$$

$$\Rightarrow A+B = A(1+2i\beta)-B(1-2i\beta)$$

$$\Leftrightarrow B = \frac{2i\beta}{2-2i\beta}A = \frac{i\beta}{1-i\beta}A$$

$$\Rightarrow F = \frac{1}{1-i\beta}A$$

We now consider the probability current

$$j(x,t) = \frac{i\hbar}{2m}\left(\psi\frac{\partial\psi^*}{\partial x} - \psi^*\frac{\partial\psi}{\partial x}\right) = \frac{\hbar}{m}\mathrm{Im}\left(\psi^*\frac{\partial\psi}{\partial x}\right)$$

To the left of the δ-function potential well, we have

$$\psi(x,t) = Ae^{ikx} + Be^{-ikx}$$

which gives

$$\psi^*\frac{\partial\psi}{\partial x} = \left(A^*e^{-ikx} + B^*e^{ikx}\right)ik\left(Ae^{ikx} - Be^{-ikx}\right)$$

$$= ik\left[|A|^2 - |B|^2 + 2i\,\mathrm{Im}\left(AB^*e^{2ikx}\right)\right]$$

and therefore

$$j(x,t) = \frac{\hbar k}{m}\left(|A|^2 - |B|^2\right)$$

This can be interpreted as

$$j = j_{in} - j_{refl} \quad \text{with} \quad \begin{cases} j_{in} = \frac{\hbar k}{m}|A|^2 \\ j_{refl} = -\frac{\hbar k}{m}|B|^2 \end{cases}$$

The reflection coefficient R is defined as $R := \frac{|j_{refl}|}{j_{in}}$

We find

$$R = \frac{|B|^2}{|A|^2} = \left|\frac{i\beta}{1-i\beta}\right|^2 = \frac{\beta^2}{1+\beta^2}$$

For the transmission we consider ψ to the right of the potential

$$\psi(x,t) = F e^{ikx}$$

$$\Rightarrow \psi^* \frac{\partial \psi}{\partial x} = F^* e^{-ikx} \cdot ik \cdot F e^{ikx} = ik|F|^2$$

$$\Rightarrow J_{trans}(x,t) = \frac{\hbar k}{m}|F|^2$$

The transmission coefficient T is defined as $T = \frac{J_{trans}}{J_{in}}$

$$\Rightarrow T = \frac{|F|^2}{|A|^2} = \left|\frac{1}{1-i\beta}\right|^2 = \frac{1}{1+\beta^2}$$